

Manish

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Skills

Languages: Python, SQL, JavaScript

Frameworks & Libraries: PyTorch, Hugging Face, LangChain, FastAPI, ReactJS, SQLAlchemy, Gradio

Core Areas: Natural Language Processing, Generative AI, Retrieval-Augmented Generation (RAG), Model Fine-tuning, Sequence Modeling, Predictive Modeling

Tools & Platforms: Ollama, Neo4j, Docker, Git, GitHub, MLflow, PySpark, Google Colab

Education

Kings Engineering College - Chennai

Sep 2022 - Apr 2026

B.Tech. in Artificial Intelligence and Data Science

CGPA: 8.3/10

Relevant Coursework: Object Oriented Programming, Databases, Discrete Maths, Statistics and Probability, Data Structures and Algorithms, Computer Networks, Exploratory Data Analysis, Machine Learning, Deep Learning, Big Data

Project Work

- **StudyGraphRAG:**
 - Built a GraphRAG-based study assistant that transforms PDF documents into a Neo4j knowledge graph, enabling graph-aware retrieval for more contextual question answering.
 - Designed an asynchronous document ingestion pipeline to extract entities, relationships, and embeddings, allowing efficient background processing of large study materials.
 - Developed FastAPI APIs for document upload, graph management, and conversational querying by integrating LLM-based answer generation with retrieved graph context.
 - Technologies: FastAPI, SQLAlchemy, React, TailwindCSS, Axios, Neo4j, GraphRAG, Gemini API.
- **SQL Easy Query + Gemma 3 Fine-Tuning:**
 - Fine-tuned Gemma 3 4B for Text-to-SQL using 4-bit QLoRA on the `b-mc2/sql-create-context` dataset and deployed the model locally using Ollama.
 - Built a full-stack AI database assistant that converts natural language into SQL queries across SQLite, PostgreSQL, MySQL, SQL Server, and Oracle using FastAPI and React.
 - Improved query reliability by implementing SQLGlot-based SQL validation, automatic query correction and AI-generated summaries of query results.
 - Technologies: PyTorch, Transformers, peft, bitsandbytes, Ollama, FastAPI, React, SQLAlchemy, SQLGlot.
- **Speech-to-Speech Conversational AI:**
 - Developed a fully local speech-to-speech AI assistant by integrating Whisper for speech recognition, Ollama for LLM inference, and Kokoro TTS with eSpeak-NG for voice synthesis.
 - Built a real-time audio pipeline to record speech, transcribe audio, generate streamed LLM responses, and synthesize natural voice output.
 - Created an interactive Gradio interface for seamless voice conversations without relying on external cloud APIs.
 - Technologies: OpenAI Whisper, Ollama, Kokoro TTS, eSpeak-NG, Gradio, PyAudio.
- **Enta – Bilingual Intent and Slot Prediction:**
 - Built a bilingual (English & Tamil) joint intent classification and slot filling model from scratch in PyTorch using the MASSIVE multilingual dataset.
 - Designed a shared Transformer encoder with task-specific heads, attention-based intent pooling, and CRF-based slot decoding for structured prediction.
 - Implemented complete training, evaluation, and inference pipelines, enabling real-time multilingual intent and entity prediction.
 - Technologies: PyTorch, Transformers, SentencePiece, streamlit.

Awards and Certificates

- **Vice President, AI Club (College):** Organized AI and machine learning seminars, mentored junior students, and conducted hands-on technical sessions.
- **Chatbot Challenge Runner-Up:** Secured 2nd place in a chatbot development competition at Sairam Institute by building a functional chatbot within 60 minutes.
- Completed Machine Learning and Data Science courses on Udemy covering supervised learning, deep learning, NLP, and basics of model deployment.